

Parth Mital

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🐙 github.com/parthmital 🌐 parthmital-portfolio.vercel.app/

Education

Vellore Institute of Technology (VIT)

B.Tech in Computer Science & Engineering (CGPA: 8.75 / 10)

2023 – 2027

Vellore, India

Experience

Concept Marketing

June 2026 – Present

Independent Software Developer (Contract)

Vadodara, India

- Modernised a legacy **ASP.NET WebForms + SQL Server ERP** into a **Next.js, TypeScript, Prisma, PostgreSQL** platform, implementing **52 APIs** across **60+ modules**.
- Executed a zero-downtime migration of **114K+ records** across **42 legacy tables** with **100% backward compatibility**, & migrated deployment to **Vercel + Supabase**, eliminating on-premise infrastructure & enabling secure cross-device access.
- Built a production ERP supporting **60+ daily users**, managing **2.8K customers**, **28K service calls**, **1.1K AMC contracts**, & **5.6K machines**.

LearnLine Edustation (IntelGrader)

June 2026

AI Engineer Intern

Remote

- Secured the internship by developing a **YOLOv8s crossout detector**; annotated **100 answer sheets** with **277 boxes** across **5 classes**, achieving **0.283 mAP@50**, **80.39% precision**, & **7.11% OCR clutter reduction**.
- Trained a **MobileNetV3-Small** document quality model on a **10K-image dataset**, compressed it by **82.5%** (**12.5 MB → 2.2 MB**), & deployed it in a **Kotlin + Jetpack Compose** Android app with **2.55 ms** on-device inference.
- Built **AI book ingestion** & **GenAI lecture generation** pipelines using **LLMs, Stable Diffusion, TTS, & T2V models**; redesigned the IntelGrader student planner frontend.

Projects

🐙 NeuroScan AI: Brain Tumour MRI Analysis System

- Conducted a comparative study of **21 deep learning architectures** for brain tumour detection, classification, & segmentation; authored **2 research papers** under review.
- Built a **3-stage AI pipeline** benchmarking **4,705 MRI scans** & **BraTS 2020**, achieving **99.59% detection**, **98.58% classification**, & **0.7655 Dice**.
- Engineered a medical imaging platform with MRI inference, 3D visualisation, volumetric analysis, PDF reporting, JWT authentication, & REST APIs.

🐙 Muse: Self-Hosted Music Streaming & Discovery Platform

- Built a **Spotify-inspired streaming platform** with a distributed **4-service architecture** using **Next.js, Fastify, PostgreSQL, & FastAPI**.
- Implemented a **Last.fm recommendation engine** with personalised ranking, **25-track queues**, radio, & up to **44 Tidal resolutions** per request.
- Built a backend with job queues, workers, JWT, Docker, & caching for **10K profiles**, **50K recommendations**, & **100K playback sessions**.

Technical Skills

Languages: C, C++, Java, Python, TypeScript, JavaScript

Coursework: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, Digital Systems, Microprocessors

Software: Full-Stack, Frontend, Backend, REST APIs, Authentication, Databases

AI/ML: Machine Learning, Deep Learning, Computer Vision, NLP, Model Training, Feature Engineering, ETL

Cloud & DevOps: AWS, Docker, Cloud Deployment, Containerization